ThoughtFlow: A Preregistered Falsification Ladder for Sparse-Cache Retention

Anonymous authors
Paper under double-blind review

Abstract

We present a preregistered falsification protocol for training-free sparse KV-retention proposals on saved reasoning-style traces. The contribution is diagnostic, not a positive method: freeze a sparse-cache quality surface, preregister each successor signal before measurement, require matched-budget wins against R-KV-like and ThinKV-like proxies, separate stopped-family rows from proxy-baseline rows, and report paired uncertainty plus oracle/headroom diagnostics. As a case study, the original anchor/recent/phase/math family is stopped, recurrence-distance utility (`rdu_topk`) clears one frozen 74-trace surface but fails stricter reproduction, prefix-surprisal utility (`psi_topk`) fails on a fresh 70-trace C2C GSM surface, and value-weighted attention contribution (`vwac_topk`) fails on a fresh 64-trace C2C SVAMP surface. The anchor/phase int8 quantization primitive passes Triton interpreter correctness tests against its CPU reference, but no real FP8, CUDA, latency, throughput, or live compression-method claim is made.

1 Contribution

The contribution is a reusable stop-rule protocol, not another sparse-cache policy. The protocol combines seven checks: no-retune frozen surfaces, pre-registered successor utilities, matched-budget proxy baselines, stopped-family/proxy-baseline separation, paired uncertainty, oracle/headroom diagnostics, and machine-checkable stop artifacts. ThoughtFlow-FP8 is the case study showing why each check matters: one plausible signal looks positive on a frozen surface, then fails to reproduce under stricter reproduction, while two later signals fail fresh-surface gates outright.

Protocol check	Purpose	Stop condition
Frozen surface	Prevent post-hoc tuning.	A successor is measured once after preregistration.
Matched budget	Avoid full-cache comparisons.	Must beat R-KV-like and ThinKV-like rows at the same keep rate.
Stopped/proxy split	Separate local family reuse from proxy-baseline robustness.	A stopped-family-only win is diagnostic, not a method claim.
Paired intervals	Avoid mean-only promotion.	Promotion requires the preregistered mean margin and paired CI high below zero.
Oracle/headroom	Show whether a useful policy exists at the budget.	Large oracle gaps block method claims.
Fresh surface	Test robustness.	Failure demotes or kills the signal.

Table 1: Falsification protocol used for each sparse-cache successor.

2 Signals Under Test

The original policy protects anchors and phase labels. The first successor adds a small recent-token reserve. The second successor protects anchors, reserves half the retained budget for recent tokens, and lets phase markers compete with math-state and high-importance reasoning tokens under a saliency bonus. Those policies are now stopped on the current saved traces. The pre-registered `rdu_topk` successor is demoted after stricter reproduction; later `psi_topk` and `vwac_topk` successors are killed by fresh-surface gates.

Policy	Keep rate	NLL	Δ NLL vs full	PPL
full context	1.000	2.101	0.000	9.7
R-KV-like	0.210	3.419	1.319	38.7
ThinKV-like	0.210	3.583	1.482	44.6
LongFlow-like	0.210	3.961	1.861	74.2
ThoughtFlow	0.210	3.961	1.861	74.2
ThoughtFlow-recent	0.210	3.562	1.461	44.2
ThoughtFlow-saliency-recent	0.210	3.434	1.333	38.1
Held-out R-KV-like	0.211	3.482	–	43.7
Held-out sweep best	0.211	3.480	–	43.7

Table 2: Case-study retained-context continuation proxy on distilgpt2 traces. Lower NLL is better. Full context is not a matched-budget baseline. The held-out sweep row is selected on 12 train traces and reported on 12 held-out traces, so its NLL is not directly averaged with the all-trace rows.

3 Hidden/KV Telemetry

We then ran a Mac-local telemetry gate using real distilgpt2 tensors rather than only text markers. The probe ranks retained tokens by attention received, final-hidden norm, key norm, value norm, or combined KV norm. This is still not sparse-KV decoding, but it checks whether the interpretable phase policy is merely preserving labels that hidden/KV saliency would not select.

Policy	Keep rate	Anchor	Phase	Math-state
ThoughtFlow	0.207	1.000	1.000	0.918
LongFlow-like	0.207	1.000	1.000	0.918
ThinKV-like	0.207	1.000	0.948	0.848
`value_norm_topk`	0.207	0.000	0.492	0.845
train-selected sweep	0.207	1.000	0.430	0.956

Table 3: Hidden/KV saliency telemetry on 24 distilgpt2 traces. The strongest real-saliency proxy by phase+math recall is `value_norm_topk`. ThoughtFlow beats it on phase recall, but LongFlow-like ties ThoughtFlow under the local importance heuristic.

Paired against `value_norm_topk`, ThoughtFlow's phase-recall delta is +0.508 with 95% CI [+0.436, +0.579]. The math-state delta is only +0.073 with 95% CI [-0.078, +0.223]. This rules out a method claim based on phase recall alone: it may be marker preservation rather than useful cache retention.

4 Sparse-Cache Case Study Gate

We next ran a CPU-only sparse-cache probe. The model processes the full prefix once, each policy prunes the returned KV cache to the same 0.20 budget, and the continuation is scored from the pruned cache. This is not a Triton, CUDA, FP8, latency, or throughput result, but it is stronger than retained-text NLL because the model actually receives a sparse cache.

Policy	Keep rate	NLL	Paired Δ vs R-KV-like
full cache	1.000	2.142	-1.296 [-1.533,-1.066]
ThoughtFlow-saliency-recent	0.210	3.372	-0.067 [-0.151,+0.011]
ThinKV-like	0.210	3.389	-0.049 [-0.192,+0.077]
ThoughtFlow-recent	0.210	3.399	-0.040 [-0.169,+0.074]
ThoughtFlow sweep best	0.210	3.436	-0.002 [-0.007,+0.000]
R-KV-like	0.210	3.438	0.000
LongFlow-like	0.210	3.588	+0.150 [-0.016,+0.300]
ThoughtFlow	0.210	3.588	+0.150 [-0.006,+0.297]

Table 4: CPU sparse-KV continuation quality on 24 distilgpt2 traces. Negative paired delta is better than R-KV-like. ThoughtFlow-saliency-recent is best in mean NLL, but it clears the strongest non-ThoughtFlow row by only 0.017 NLL and remains below the pre-registered 0.03 promotion margin.

A bounded sparse-cache sweep over 24 nearby anchor/recent/phase/math configurations selects `tf_sparse_r0.55_p0.05_m0.12_a2` on the 12 train traces. On the 12 held-out traces it gets NLL 3.340 versus ThinKV-like 3.385 and R-KV-like 3.420. This clears the mean 0.03 margin, and the paired delta versus R-KV-like is -0.080 with 95% CI [-0.152,-0.014]. However, the paired delta versus ThinKV-like is -0.045 with 95% CI [-0.226,+0.182]. The fixed ThoughtFlow-saliency-recent incumbent has lower held-out mean NLL, 3.304, but its paired CIs also cross zero. This is a promising falsification candidate, not a method result.

We then froze `ThoughtFlow-saliency-recent` and `tf_sparse_r0.55_p0.05_m0.12_a2` and ran a larger 74-trace sparse cache slice with no retuning. ThinKV-like is the best compressed row among the stopped family comparison at NLL 3.900. The frozen sparse ThoughtFlow row gets 3.908, with paired delta -0.031 versus R-KV-like and 95% CI [-0.078,+0.020], but +0.008 versus ThinKV-like with 95% CI [-0.060,+0.085]. ThoughtFlow-saliency-recent gets 3.920. The larger no-retuning gate therefore stops this policy family.

After that stop decision, we pre-registered `rdu_topk`. It computes recurrence-distance utility from full-prefix prefill attentions using lag buckets [8,15], [16,31], [32,63], and [64,∞], then keeps the top budget by utility with ties broken by lower position. It does not use continuation tokens, trace labels, recent reserves, anchors, phase markers, or math-state bonuses. After `rdu_topk` failed reproduction, we pre-registered `psi_topk` (high self-surprisal prefix tokens) and `vwac_topk` (future attention weighted by cached value-vector norm).

5 Evidence Ladder

The independent saved-trace check rules out claiming `rdu_topk` as a robust method on the available surfaces. A coarse trace-level decomposition explains where the reproduction fails: `rdu_topk` is +0.213 NLL worse than R-KV-like on long-prefix/high-RDU-density rows but -0.049 better on short-prefix/low-density rows. The large-regression buckets are exactly the buckets a robust policy would need to handle before any long-context claim.

Prefix-surprisal successor. A later consumed successor tested whether retaining high self-surprisal prefix tokens could preserve information that is hard to reconstruct. The signal uses only prefill logits, not continuation loss, trace labels, recent reserves, or RDU lag buckets. On a fresh 70-trace C2C GSM saved-trace surface, `psi_topk` fails decisively: ThinKV-like is best compressed at NLL 3.906, R-KV-like reaches 3.960, and `psi_topk` reaches 7.899. Paired deltas are +3.939 [+3.720,+4.144] versus R-KV-like and +3.994 [+3.773,+4.205] versus ThinKV-like, where positive is worse for `psi_topk`. This rules out this exact signal in the current Mac sparse-cache harness.

Value-weighted attention successor. The next consumed successor tested a cache-state utility: future prefix attention to a token multiplied by the token's cached value-vector norm. On a fresh 64-trace C2C SVAMP surface, `vwac_topk` also fails. R-KV-like is best com-

Gate	Readout	Decision
Synthetic retention	Anchor/phase labels can be protected at matched keep rate.	Intuition only.
Retained-text NLL	Original ThoughtFlow rows do not beat the strongest proxy.	Weakened.
Hidden/KV telemetry	Phase recall improves, but math-state impact is not stable and LongFlow-like ties the heuristic.	Diagnostic.
CPU sparse-cache probe	A tuned sparse ThoughtFlow row is promising in mean but not robust against ThinKV-like uncertainty.	Not promoted.
Frozen 74-trace gate	`rdu_topk` beats ThinKV-like and R-KV-like on the first frozen surface.	Candidate only.
Same-slice rerun	Cached result reproduces exactly on the deterministic Mac stack.	Repro bookkeeping.
Alternate surface	Cross-family rows still lose, but a stopped same-family row beats `rdu_topk` by 0.006 NLL.	Weakened.
Independent saved traces	R-KV-like is best compressed at NLL 3.981; `rdu_topk` reaches 4.014.	Killed.
Prefix-surprisal fresh surface	`psi_topk` reaches NLL 7.899 versus ThinKV-like 3.906 and R-KV-like 3.960.	Killed.
Value-weighted attention fresh surface	`vwac_topk` reaches NLL 4.336 versus R-KV-like 4.096 and ThinKV-like 4.162.	Killed.

Table 5: ThoughtFlow-FP8 evidence ladder. The first-surface pass is shown as one step that failed to reproduce, not as the headline result.

Policy	Traces	Keep rate	NLL	Paired Δ vs ThinKV-like
full cache	74	1.000	2.848	-1.052 [-1.199,-0.919]
`rdu_topk`	74	0.213	3.779	-0.121 [-0.211,-0.037]
ThinKV-like	74	0.213	3.900	0.000
frozen sparse ThoughtFlow	74	0.213	3.908	+0.008 [-0.060,+0.085]
ThoughtFlow-saliency-recent	74	0.213	3.920	+0.020 [-0.030,+0.074]
R-KV-like	74	0.214	3.939	+0.039 [-0.015,+0.099]
LongFlow-like	74	0.213	4.158	+0.258 [+0.178,+0.337]

Table 6: Historical one-shot frozen sparse-cache successor evaluation. `rdu_topk` also beats R-KV-like by -0.160 NLL with 95% CI [-0.264,-0.050] on this first surface, but later alternate-surface and independent-trace gates fail to reproduce the promotion.

pressed at NLL 4.096, ThinKV-like reaches 4.162, and `vwac_topk` reaches 4.336. Paired deltas are +0.241 [+0.030,+0.530] versus R-KV-like and +0.174 [+0.002,+0.392] versus ThinKV-like.

6 Claim Boundary

Limitations. The ladder is demonstrated on Mac-local saved traces and CPU sparse-cache scoring, not native serving. The trace families are useful falsification surfaces, but they do not establish broad task coverage. The stopped signals also do not rule out future sparse-cache utilities; they only rule out the pre-registered policies under the reported surfaces and budgets. The scored surfaces use `distilgpt2` saved continuations and saved prefix/KV telemetry as falsification fixtures; they are not reasoning-model benchmarks and not faithful external implementations of ThinKV, R-KV, LongFlow, LazyEviction, ForesightKV, PM-KVQ, or KVQuant.

`rdu_topk` check	Readout	Decision
First frozen surface	74 traces, NLL 3.779 versus ThinKV-like 3.900.	First-surface pass.
Same-slice rerun	Exact deterministic reproduction; 0.145 NLL above compressed oracle.	Bookkeeping only.
Alternate surface	Same rule, NLL 3.594; a stopped same-family row reaches 3.588.	Weakened.
Independent traces	89 SVAMP traces; R-KV-like 3.981, `rdu_topk` 4.014.	Killed.
Failure bucket	+0.213 NLL on long-prefix/high-RDU-density rows.	Mechanism target.

Table 7: `rdu_topk` demotion ladder. The first-surface pass is preserved as evidence that the protocol catches tempting positives, not as a live method claim.

Claimed	Not claimed
Preregistered sparse-cache falsification ladder.	State-of-the-art KV compression.
Matched-budget Mac sparse-cache scoring.	Native FP8 serving quality.
Stopped-family/proxy-baseline and fresh-surface stop rules.	CUDA kernel speed, latency, throughput, or HBM savings.
Triton-interpreter parity for an int8 anchor/phase primitive.	A live deployable compression method.

Table 8: Boundary between the workshop claim and the stopped systems-method target.

7 Stop Rule and Reopen Gate

The branch is now demoted to a diagnostic. The stopped anchor/recent/phase/math policy family remains ruled out as a tunable branch on this trace set, `rdu_topk` fails reproduction, and `psi_topk`/`vwac_topk` fail their fresh-surface gates. A paper-ready method claim now requires a genuinely new pre-registered utility signal evaluated once on a fresh/larger frozen surface with strict stopped-family, proxy-baseline, and oracle/headroom reporting. The Triton interpreter gate is no longer the blocker for the quantization scaffold; method evidence is. This means no real FP8, CUDA, latency, or throughput result is claimed.

8 Related Work and Scope

This draft intentionally does not claim a new state-of-the-art KV-compression method. ThinKV, LongFlow, R-KV/R-KVHash-style retention, LazyEviction/ForesightKV-style future-use ideas, PM-KVQ, and KVQuant already occupy much of the sparse-cache and quantized-cache design space ThinKV (2026); LongFlow (2026); R-KV (2025); LazyEviction (2025); ForesightKV (2026); PM-KVQ (2025); KVQuant (2024). Our contribution is narrower: a Mac-local falsification ladder that shows how plausible interpretable retention signals can clear one frozen surface, fail stricter reproduction, or collapse on fresh surfaces. The project name retains “FP8” because that was the original systems target, but the current artifacts only validate CPU sparse-cache scoring and an int8/Triton-interpreter reference primitive; they do not measure FP8 numerical drift or native FP8 serving.

9 Reproducibility

All reported gates use tracked scripts under `experimental/thoughtflow_fp8/phase2/` and the repo-local `./venv_arm64`. The frozen sparse-cache probe, same-slice rerun, alternate-surface rerun, independent saved-trace rerun, and fresh-surface successor checks each write Markdown and JSON artifacts with promotion criteria, paired intervals, oracle/headroom diagnostics, and no-retuning decisions. Several older JSON artifacts do not yet record full model revision and input-hash provenance, so the current package should be read as a machine-checked falsification note rather than a full artifact-evaluation release. The stable local correctness command is: `./venv_arm64/bin/python -m pytest experimental/thoughtflow_fp8/phase2/tests experimental/thoughtflow_fp8/phase4/tests -q -rs`.

10 Artifacts

Key artifacts are the current decision manifest, reviewer pack, workshop shell, README/progress handoff, RDU independent reproduction check, PSI/VWAC fresh-surface checks, and repo-local Triton interpreter note: `experimental/thoughtflow_fp8/phase2/current_decision_manifest_20260506.md`, `experimental/thoughtflow_fp8/paper/reviewer_pack.md`, `experimental/thoughtflow_fp8/paper/colm_workshop_shell.md`, `experimental/thoughtflow_fp8/README.md`, `experimental/thoughtflow_fp8/progress.md`, `experimental/thoughtflow_fp8/phase2/rdu_independent_trace_reproduction_check.md`, `experimental/thoughtflow_fp8/phase2/psi_fresh_sparse_cache_check.md`, `experimental/thoughtflow_fp8/phase2/vwac_fresh_sparse_cache_check.md`, and `experimental/thoughtflow_fp8/phase4/triton_interpreter_note_20260506.md`.

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